

Worksheet · Section 3.14–3.16

Libraries, Random Values & Simulations



The Big Picture

Libraries give you ready-made procedures; RANDOM adds unpredictability; simulations use both to model the real world. Practice reading documentation, finding possible outputs, and evaluating simulations.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|-------------------------------|---|
| _____ 1. Library | A. An abstraction that models a real-world process |
| _____ 2. API | B. Includes the endpoints of a range |
| _____ 3. Documentation | C. A prediction that can be tested |
| _____ 4. RANDOM(a, b) | D. A specification of how a library's procedures behave and are used |
| _____ 5. Inclusive | E. Returns a random integer from a to b, including both |
| _____ 6. Probability | F. Explains a procedure's name, parameters, and return value |
| _____ 7. Simulation | G. Something a model treats as true to simplify reality |
| _____ 8. Variability | H. How likely an outcome is, from 0 to 1 |
| _____ 9. Assumption | I. A collection of procedures you can use in new programs |
| _____ 10. Hypothesis | J. Natural randomness in real-world events |

Part 2 · Fill It In

Use each word from the word bank once.

bias

simulation

library

variability

inclusive

API

documentation

1/6

1. A collection of procedures you can use in your programs is a _____.
2. An _____ describes how a library's procedures behave and are used.
3. You learn a procedure's parameters and return value from its _____.
4. RANDOM(1, 6) is _____, so it can return 1 or 6.
5. The probability that RANDOM(1, 6) returns 4 is _____.
6. Modeling a hurricane's path on a computer is a _____.
7. Random numbers in simulations model real-world _____.
8. Leaving important factors out of a simulation can create _____.

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Practice

Part 3 · Possible or Not?

Question	Answer
Can <code>RANDOM(3, 7)</code> return 7?	_____
How many values can <code>RANDOM(0, 4)</code> return?	_____
Smallest and largest of <code>RANDOM(1, 4) + RANDOM(1, 4)</code>	_____
All possible values of <code>RANDOM(1, 3) * 10</code>	_____
Probability that <code>RANDOM(1, 10) ≤ 4</code>	_____

Part 4 · Read the Docs

sortList(list, order) — returns a NEW list with the elements sorted. *order* is "asc" or "desc". The original list is not changed.

- Parameters: _____
- `sortList([3, 1, 2], "desc")` returns _____
- After `x ← [5, 2]` and `y ← sortList(x, "asc")`, `x` is _____
- What you DON'T need to know: _____

Part 5 · Build a Simulation

Simulate 100 free throws by a player who makes 70% of shots.

```
made ← 0
REPEAT _____ TIMES
{
  IF (RANDOM(1, _____) ≤ _____)
  { made ← made + 1 }
}
DISPLAY(made)
```

Expected result: about _____. Exactly that every time? _____

Part 6 · Evaluate the Model

A fire-drill simulation assumes every student walks at the same speed and every hallway is always clear. Name two simplifications, how each could bias the results, and one improvement.



Think:

`RANDOM(a, b)` has $(b - a + 1)$ possible values. For a $P\%$ chance, use `RANDOM(1, 100) ≤ P` (or a smaller equivalent like `RANDOM(1, 10) ≤ 7` for 70%).

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. Which value can `RANDOM(5, 9)` NOT return?

- A. 10
- B. 7
- C. 5
- D. 9

★★ On Level

2. Which expression picks a random element from the list `colors`?

- A. `RANDOM(colors)`
- B. `colors[RANDOM(0, LENGTH(colors))]`
- C. `colors[RANDOM(1, LENGTH(colors))]`
- D. `colors[RANDOM(1, LENGTH(colors) + 1)]`

★★ On Level

3. What is the main benefit of using a library?

- A. It makes programs undecidable
- B. It removes the need for documentation
- C. It reuses tested code and saves development time
- D. It guarantees there are no bugs

★★★ Exam Hard

4. What is the probability that this code displays 3?

```
x ← RANDOM(1, 2)
y ← RANDOM(1, 2)
DISPLAY(x + y)
```

- A. $1/2$
- B. $1/4$
- C. $3/4$
- D. $1/3$

★★★ Exam Hard

5. Which TWO statements about simulations are true? (*Select two.*)

- A. Running more trials makes the average results more reliable.
- B. They can test scenarios too dangerous to try in real life.
- C. They cannot include randomness.
- D. They always produce exactly the real-world result.

Part 8 · Written Response

Design a simulation to decide whether a school should add a second lunch line. Describe what is random, what you would measure, and one assumption you'd make.
